Computer Networks

计算机网络（CST2398）

2025-2026学年第二学期（26-Spring）

计算机科学与技术学院 王洁

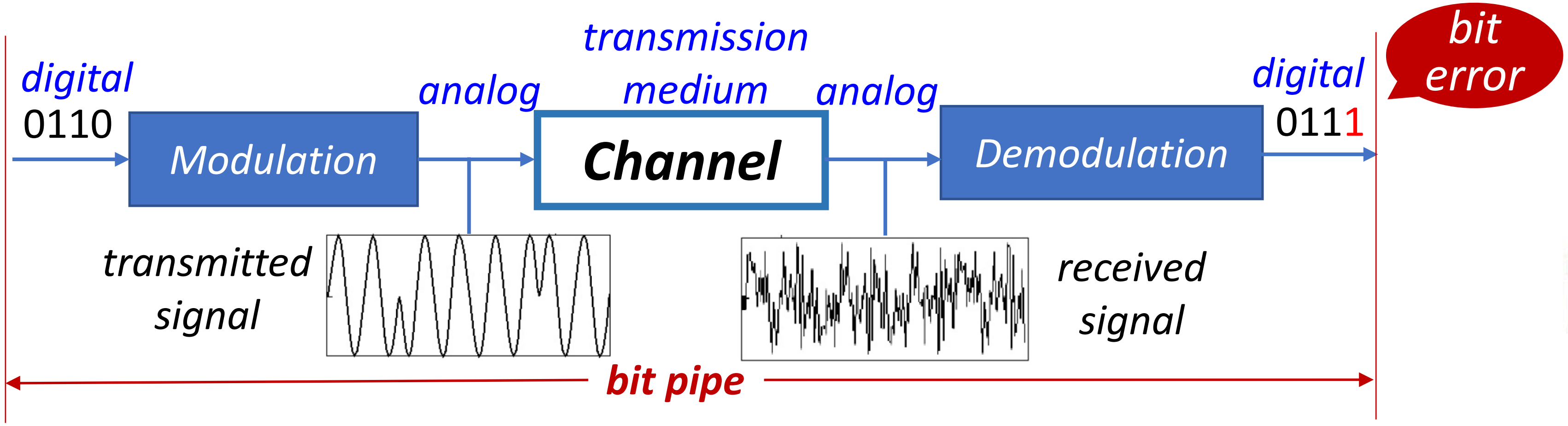
wangjie_tongji@tongji.edu.cn

02

PHY Layer

Contents of this Chapter

- ***Bandwidth (带宽)*** *and* ***Data Rate***
- *Modulation (调制) of a Signal (ASK, PSK, QPSK, QAM)*
- *Medium and Transmission (传输)*
- ***Multiplexing (复用)*** *(FDMA, TDMA, CDMA)*



Application

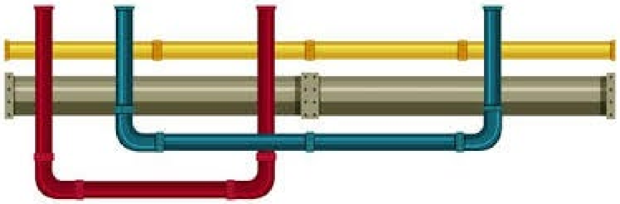
Transport

Network

Link

Physical

Providing a ***bit pipe***



2.1

BW and Data Rate

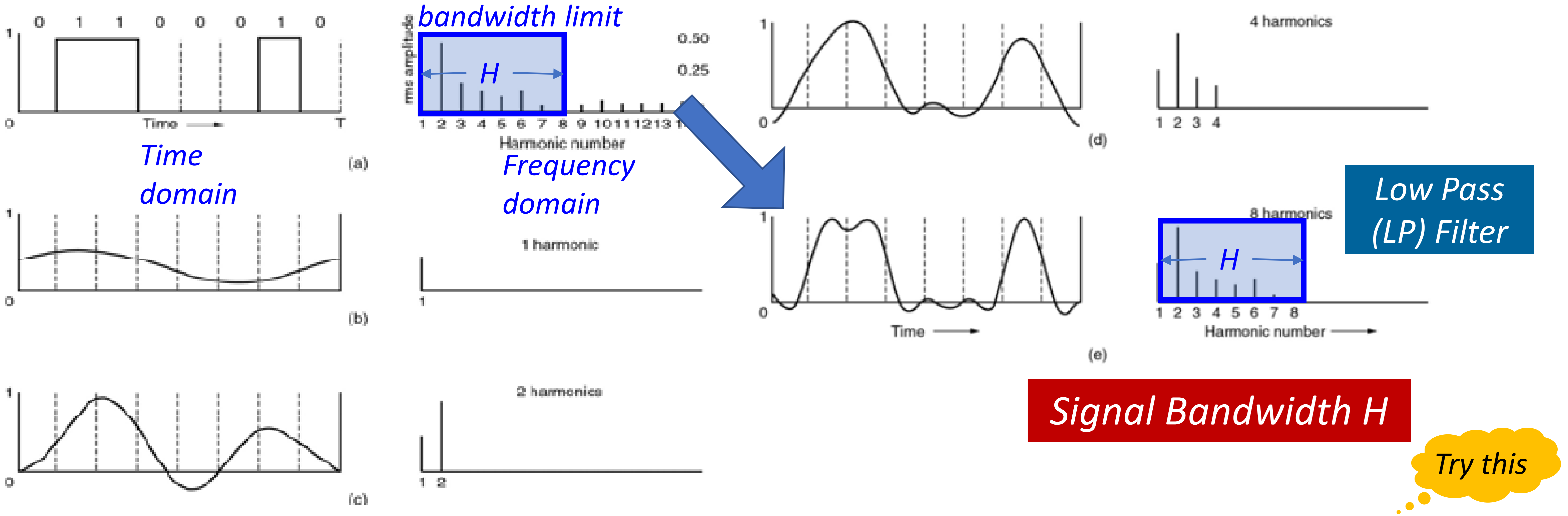
Frequency and Bandwidth-limited Signal

Representing a periodic signal

$$g(t) = \frac{c}{2} + \sum_{n=1}^{\infty} a_n \sin(2\pi n f t) + \sum_{n=1}^{\infty} b_n \cos(2\pi n f t)$$

Harmonics 谐波

Non-periodic signal
Fourier Transform 傅立叶变换
Frequency domain



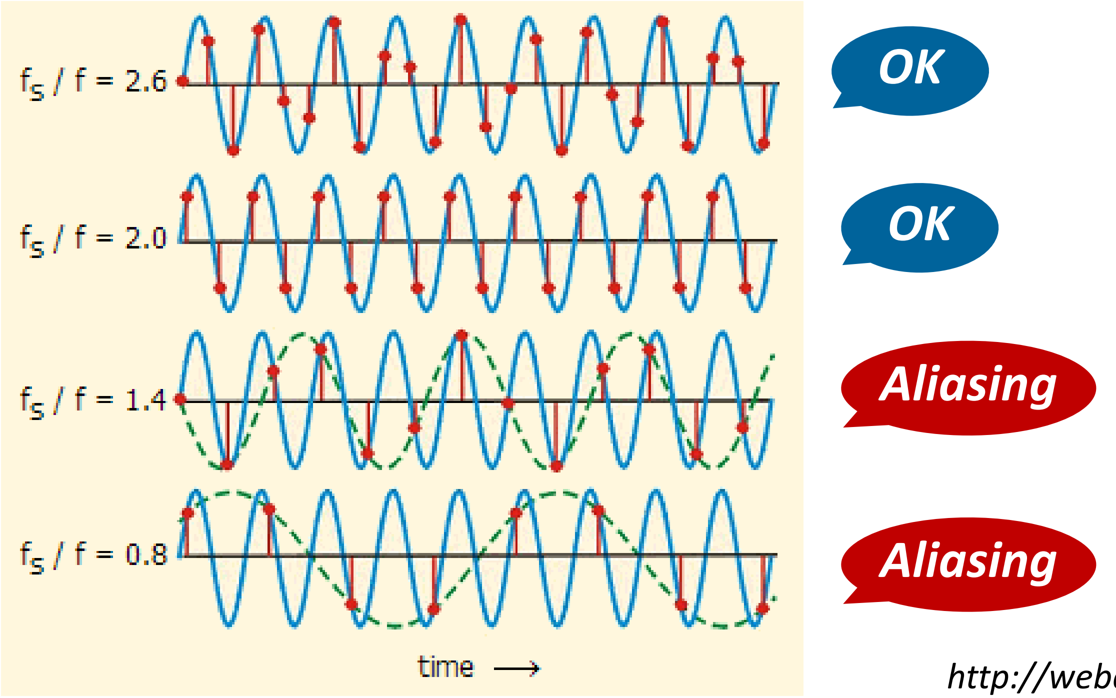
http://webapps.chem.uoa.gr/efs/applets/AppletFourier/Appl_Fourier2.html

Nyquist Sampling: restoring the signal

- ***For a signal with bandwidth H,***

Nyquist Sampling Rule

  - *To reconstruct the signal (fully), we need 2H samples per second.*
  - *Or (if sampling frequency < 2H), there will be **aliasing(混叠)**.*



Try this

http://webapps.chem.uoa.gr/efs/applets/AppletNyquist/Appl_Nyquist2.html

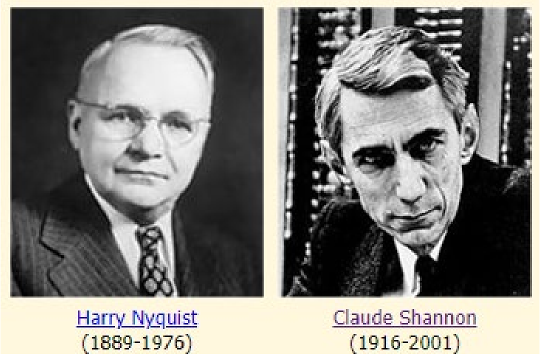
Channel Bandwidth and Data Rate

- ***For a channel with bandwidth BW,***

Due to Quantization

  - *Noiseless channel: V is the # of levels of the signal*

    max datarate $R_{max} = 2\ BW\ log_2 V$ *bps* **Nyquist Theorem**



  - *Noisy channel: S is the signal strength, N is the noise strength/power*

    SNR

    max datarate $C = BW\ log_2(1 + \frac{S}{N})$ *bps* **Shannon Theorem**

A Mathematical Theory of Communication, 1948

信道容量 **Shannon Capacity** *The maximum # of bits a channel of BW wide allows in one second.*

decibels

SNR *(signal to noise ratio, 信噪比)* = $\frac{S}{N}$ ***SNR (dB)*** = $10\log_{10}\frac{S}{N}$

SNR =3 dB, S/N = ?

2.2

Modulation

Modulation: Digital -> Analog

- ***Modulation Process**: switching (also called keying) the **amplitude(幅度)**, **frequency(频率)**, or **phase(相位)** of the carrier (载波) in accordance with the information (binary digits, 0s and 1s)*

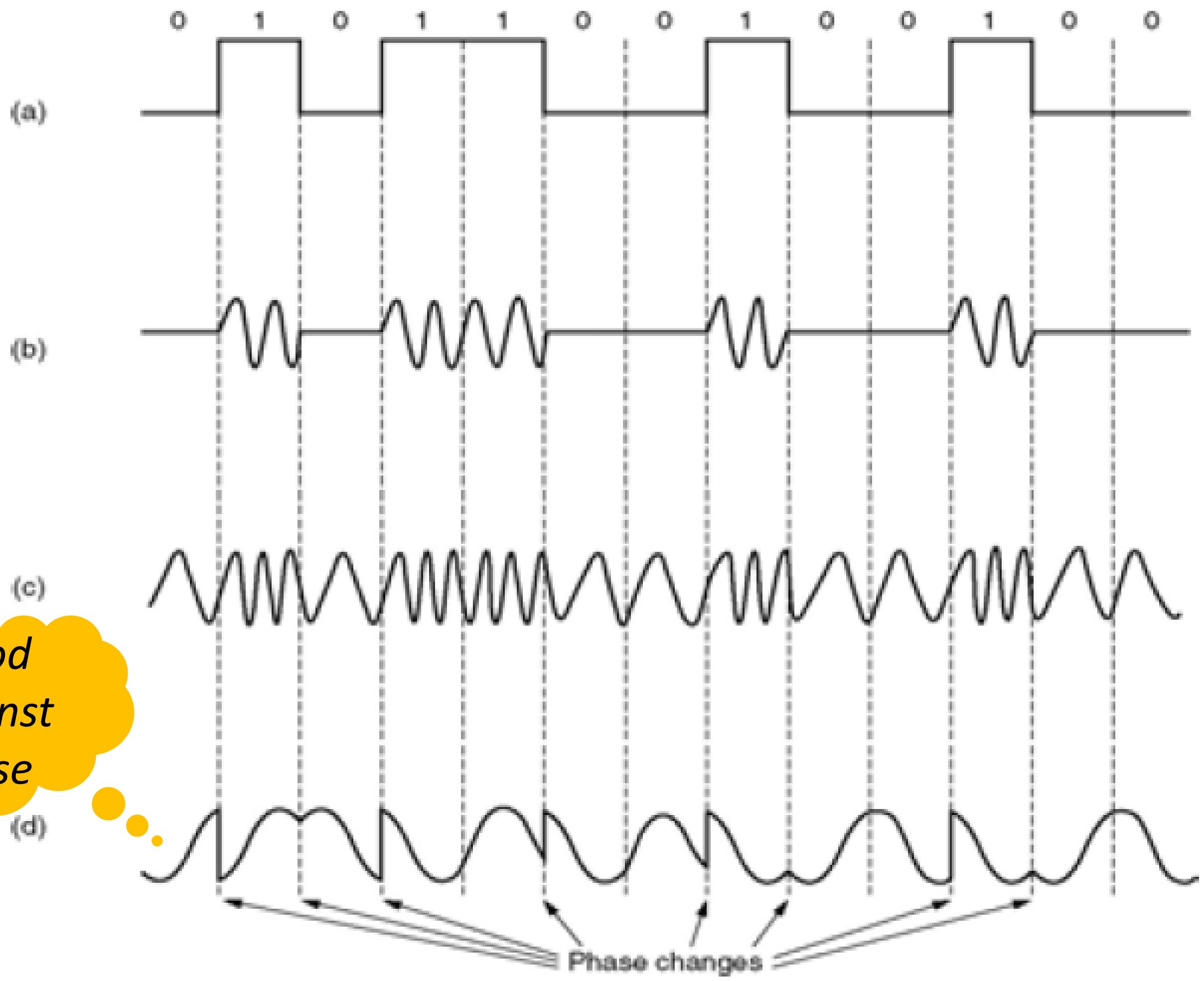
Digital: binary digits that carries information

⬇

Analog Signal: $s(t) = A\cos(2\pi f t + \theta)$

- ***Why**: channel/medium permits bandpass (BW-limited) signal*

Types of Modulation



good against noise

$$s(t) = A\,cos(2\pi f t + \theta)$$

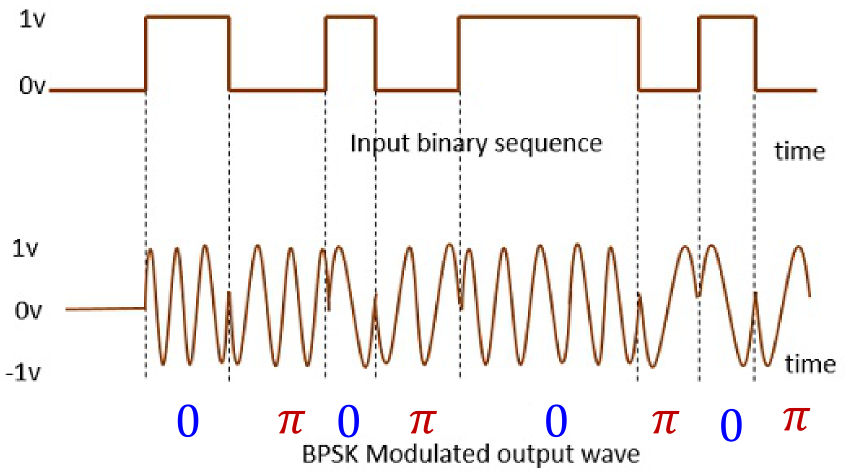
Or a combination

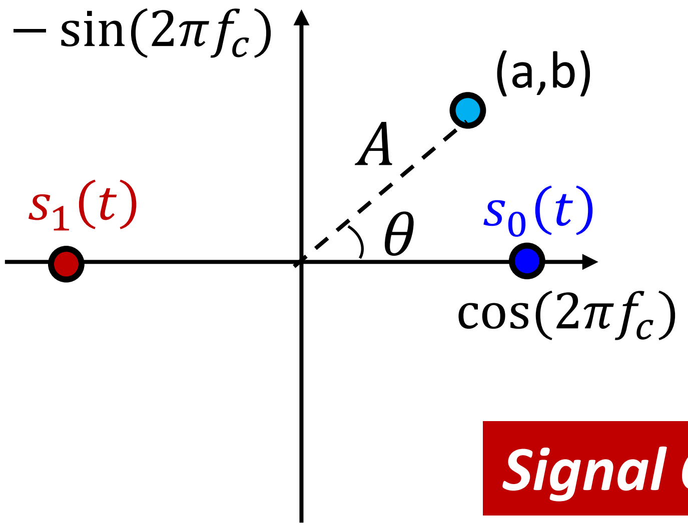
a. A binary signal
b. Amplitude Modulation (AM) or **Amplitude Shifted Keying (ASK)**
c. Frequency Modulation (FM) or **Frequency Shifted Keying (FSK)**
d. Phase Modulation (PM) or **Phase Shifted Keying (PSK)**

Binary Phase Shift Keying (BPSK)

- **BPSK**: *A binary digit (0 or 1) is mapped to a (usually high frequency f_c) carrier sinusoidal wave of a given phase.*
  - *Binary digit 0:* $s_0(t) = A\cos(2\pi f_c t) = A\cos(2\pi f_c t + 0)$
  - *Binary digit 1:* $s_1(t) = -A\cos(2\pi f_c t) = A\cos(2\pi f_c t + \pi)$

π: 180°





What are the coordinates of $s_0(t)$ and $s_1(t)$?

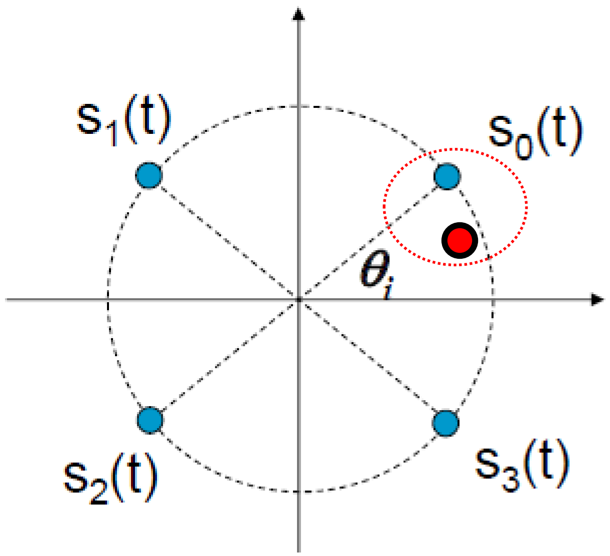
Signal Constellation

More PSK schemes

- ***More than binary (2)!***

Why?
Data rate

- ***4 -> Q(uadratic) PSK***
  - *signal $s_i(t)$ -> a symbol*



2 bits

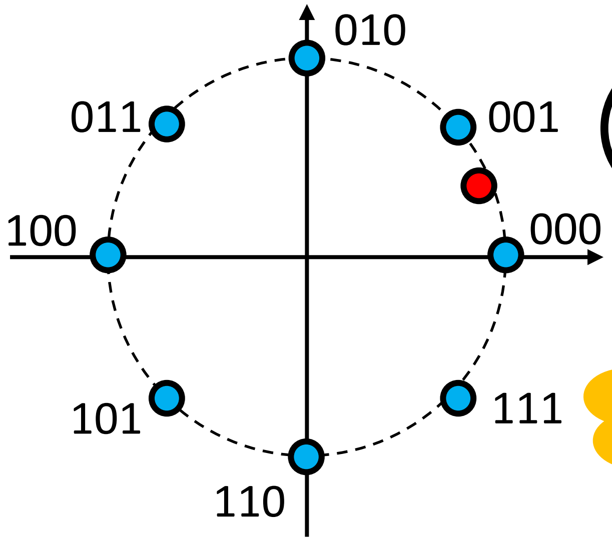
$s_0(t)$ -- 00

$s_1(t)$ -- 01

$s_2(t)$ -- 10

$s_3(t)$ -- 11

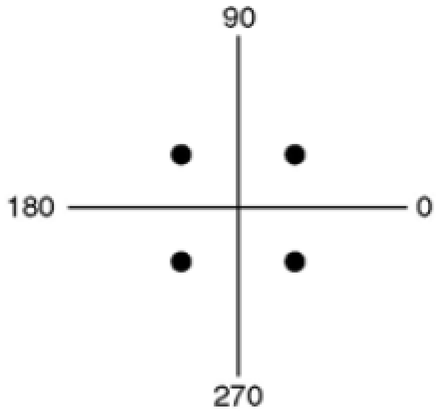
- **8PSK**
  - *signal $s_i(t)$ -> a symbol of 3 bits*



But …
Demod is
more difficult

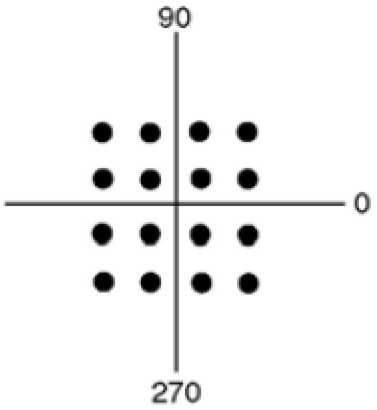
Real-world Modem: QAM

- **QAM: Quadratic Amplitude Modulation**
  - *Changing: Phase and Amplitude*



(a)

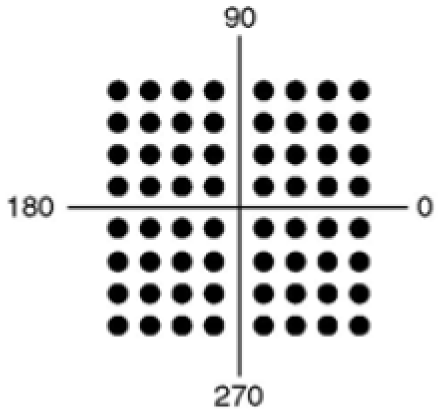
QPSK



(b)

QAM-16

4 bits/symbol



(c)

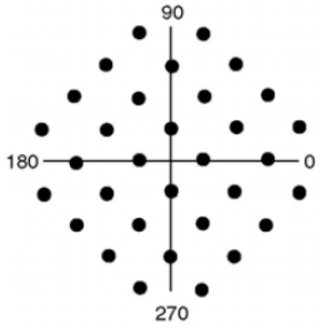
QAM-64

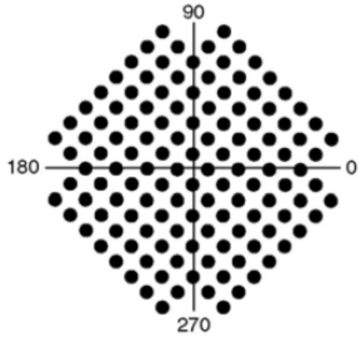
6 bits/symbol

- ***Modem sampling rate: 2400 Baud*** *1 Baud = 1 sample/s*

Data rate:

- *QAM-32*: 9.6 kbps*
- *QAM-128*: 14.4 kbps*





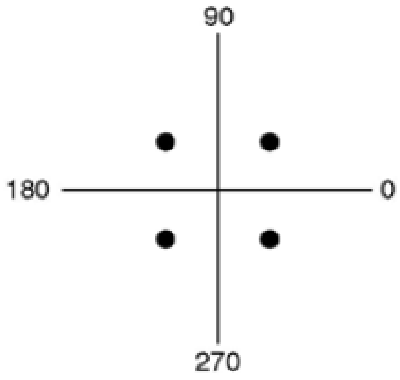
**: not the max data rate*

Ex.2.1 Data Rate, Bandwidth, Channel Capacity

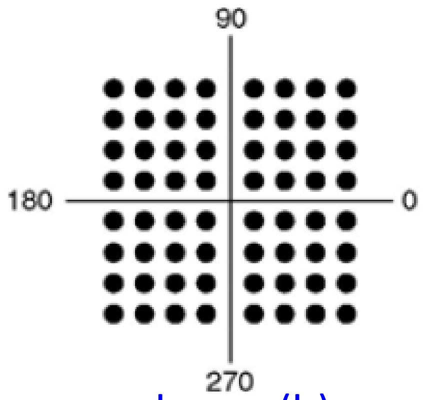
Consider two modulation schemes represented by the following diagrams:

- *Baud Rate (波特率) : 8 Baud/s (i.e. 8000 symbols/s)*
- *Each symbol is represented by one of these points*

How many bits?



scheme (a)



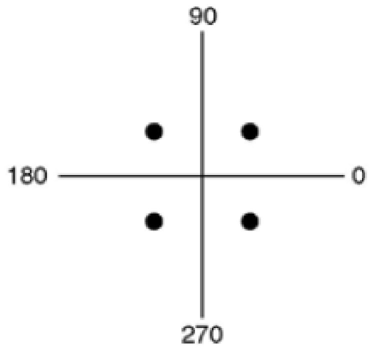
scheme (b)

- ***Questions: For each scheme,***
  1) *What kind of modulation is being used here?*
  2) *What is the data rate (in bps) for this modulated signal?*
  3) *For a noisy channel with Bandwidth 10 KHz, what is the minimum SNR to achieve this data rate?*

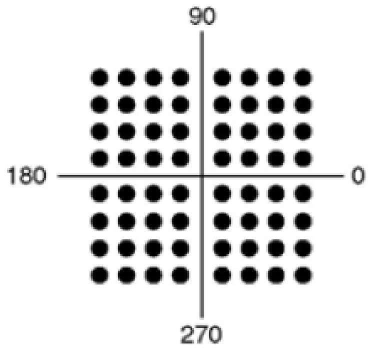
Ex.2.1 Data Rate, Bandwidth, Channel Capacity

Consider two modulation schemes represented by the following diagrams:

- *Baud Rate (波特率) : 8 Baud/s (i.e. 8000 symbols/s)*
- *Each symbol is represented by one of these points*



scheme (a)



scheme (b)

Scheme (a)

1) *QPSK*
2) *Signal data rate: 8K sample/s * 2 bit/sample = 16 kbps,*
3) *Channel BW = 10 KHz, channel capacity (max possible data rate)*

$$C = BW\ log_2(1+\frac{S}{N}) \geq 16\ kbps$$

So SNR $\geq 2^{1.6} - 1 \simeq 2.03$

Scheme (b)

1) *QAM-64*
2) *Data rate: 8K sample/s * 6 bit/sample = 48 kbps*
3) *SNR* $\geq 2^{4.8} - 1 \simeq 26.86$

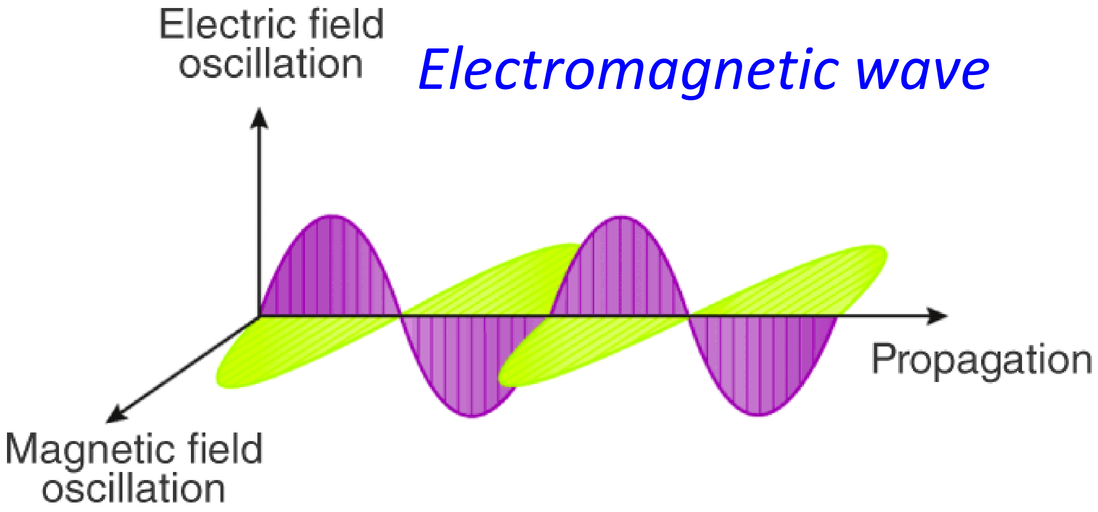
Good channel condition -> higher modulation -> higher data rate

2.3

Transmission

Transmission/Physical Media

- ***Media***: *where analog signal (that carries information) propagates*

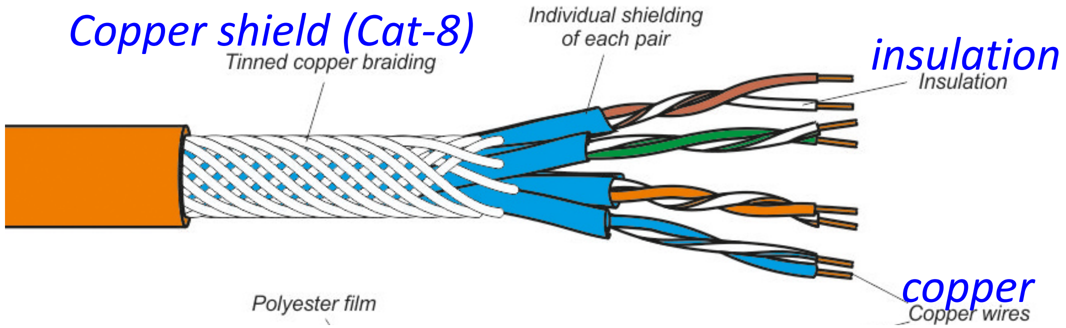


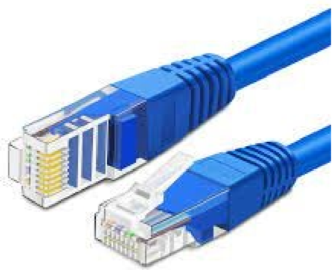
- ***Guided media:*** *signal propagates along a direction, usually solid media,*
  e.g. copper (wire), fiber, co-ax cable
- ***Unguided media:*** *signal propagates freely,*
  e.g. air, vacuum

- ***Physical link***: *the media between the sender and receiver*

Guided Media (1/2)

- ***Twisted Pair* (双绞线)**



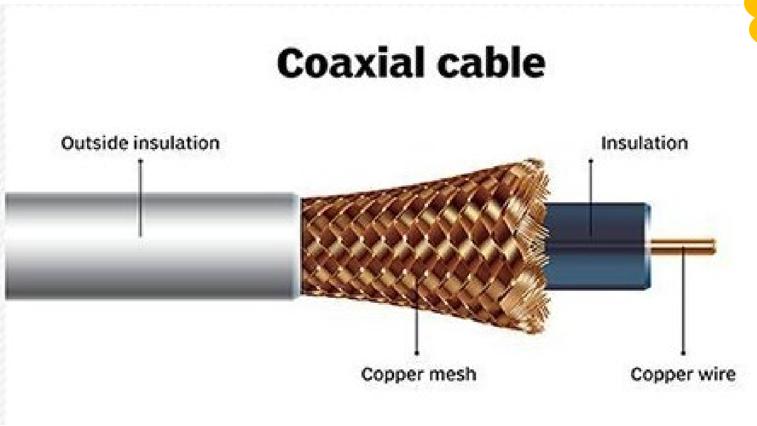


< 100m

- ***Cat-3:*** *phone wire*
- ***Cat-5:*** *100 Mbps Ethernet*
- ***Cat-5e:*** *1 Gbps Ethernet*
- ***Cat-6:*** *10 Gbps Ethernet*
- ***Cat-8****: 40 Gbps data-center*

- ***Coaxial Cable* (同轴电缆)**

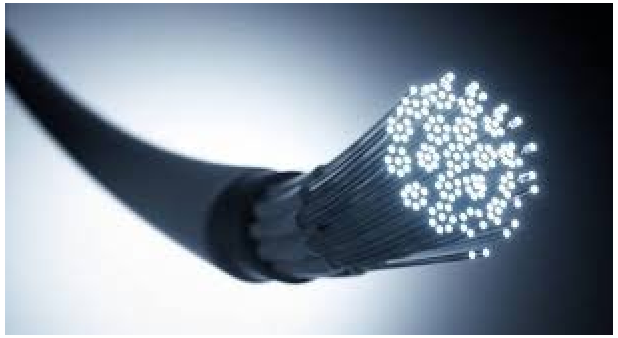
Co-ax

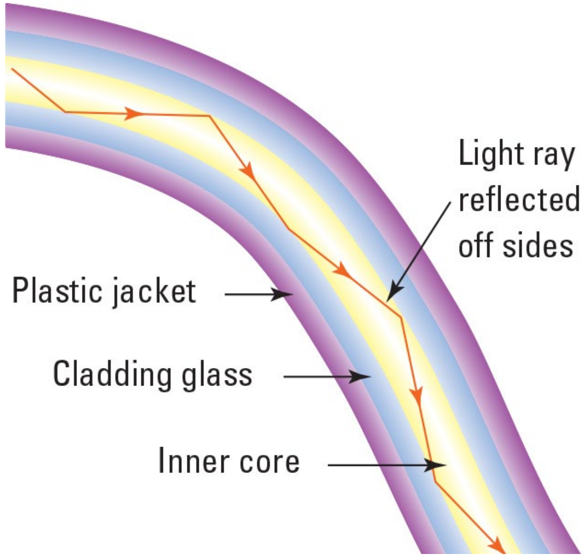


- *Cable broadband Internet*
- *Aircraft, military, medical*
- *TV, video equipment*

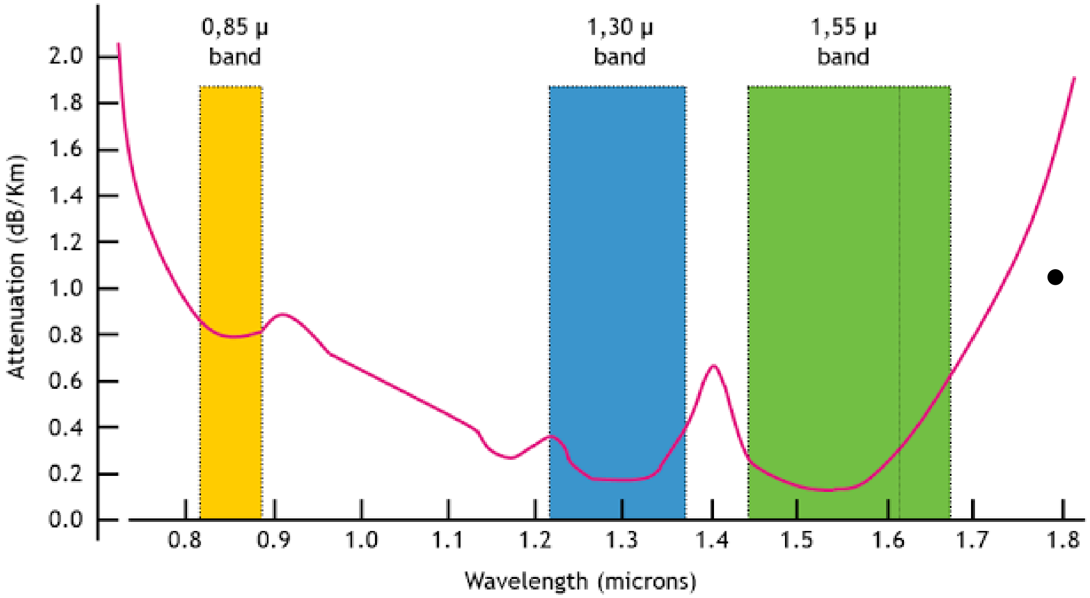
Guided Media (2/2)

- ***Fiber Optic* cable (光纤):** *glass fiber carrying light pulses*





- ***High speed:*** *10-100 Gbps*
- ***Low error rate:*** *< 10^{-12} immune to electromagnetic noise*
- ***Long distance:*** *> 100 Km*



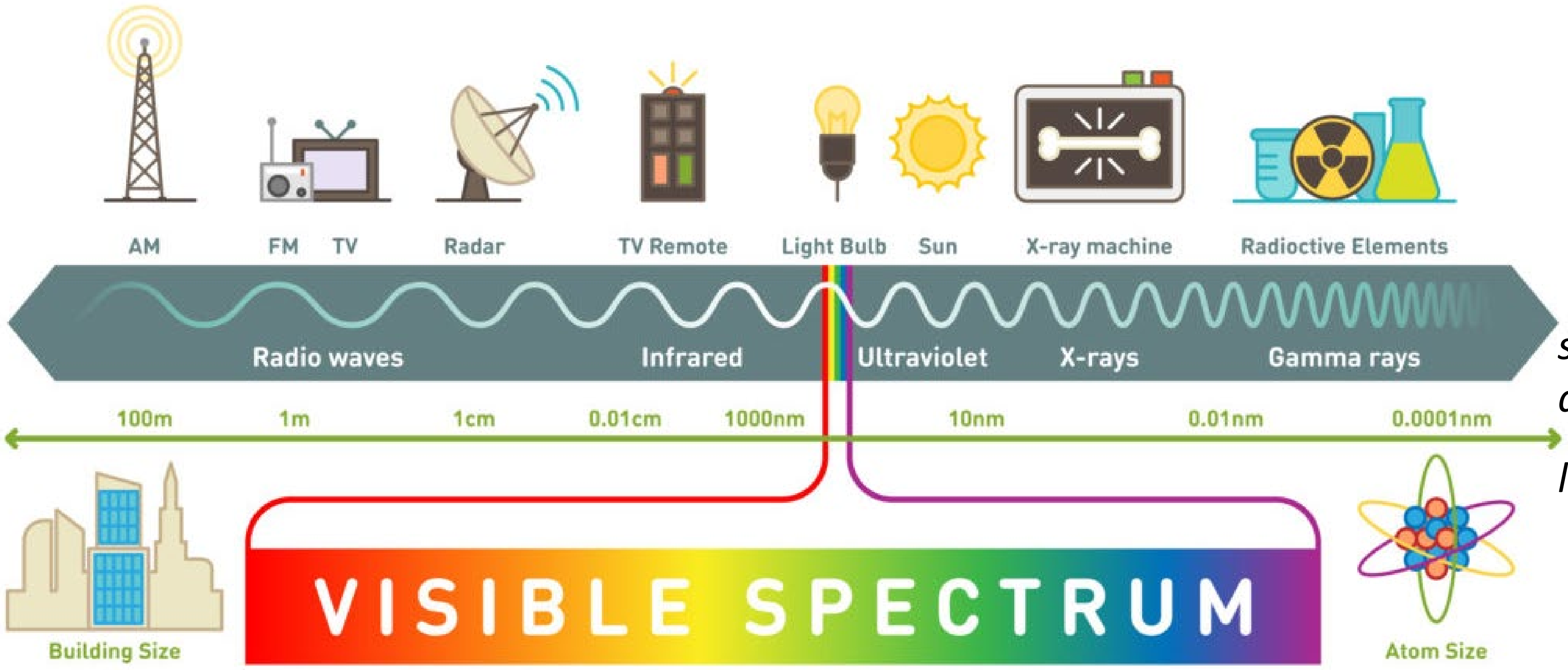
- ***Multi-mode:***
  - *Low cost*
  - *LED*
  - *short distance (Kms)*
- ***Single-mode:***
  - *Expensive*
  - *Laser*
  - *long distance (100 Km)*

The Electromagnetic Spectrum

- ***A valuable (and expensive) resource***

$\lambda \cdot f = c$

c: speed of light in the vacuum



wavelength λ (m)

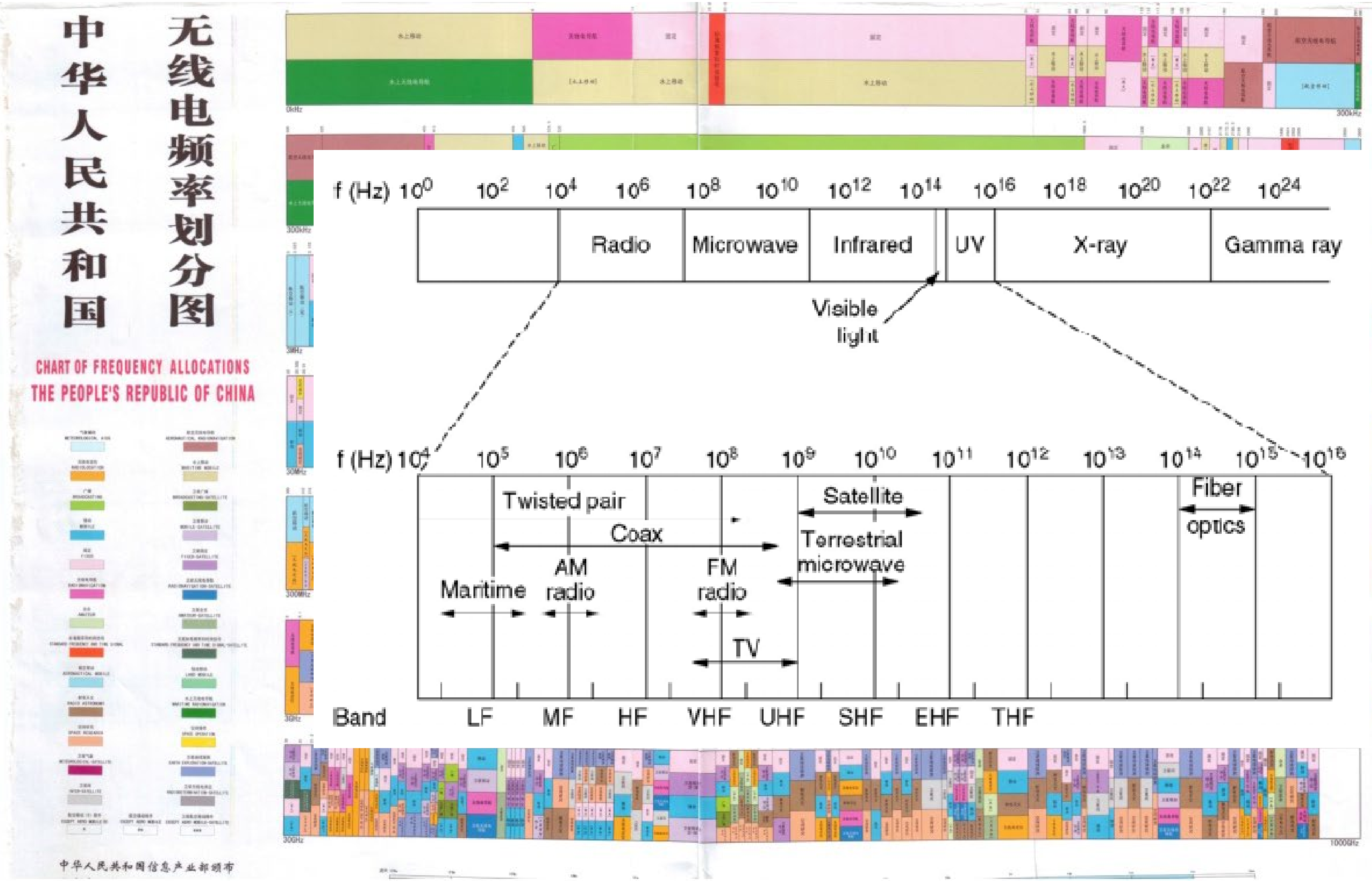
stronger attenuation

larger BW

frequency f (Hz) 10^6 10^8 10^{10} 10^{12} 10^{16} 10^{18} 10^{21} 10^{23}

MHz *GHz* *THz*

Wavelength of 2.4 GHz WiFi?



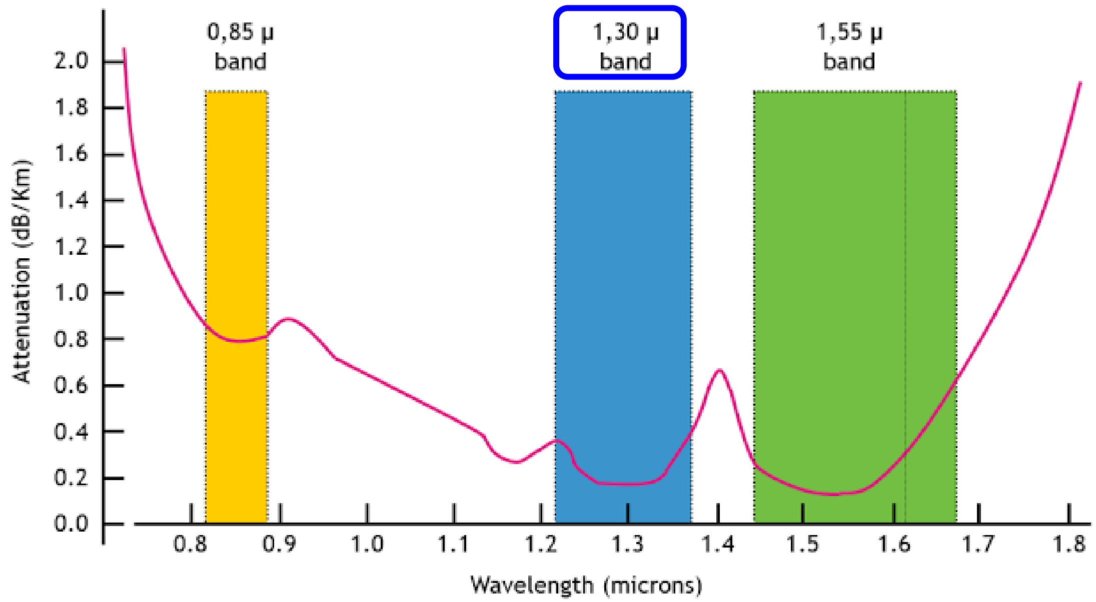
Very crowded

Ex.2.2 Frequency, Wavelength, and BW

Consider the 1.3-micron band:

- *Central frequency wavelength*
  $$\lambda = 1.3\ \mu m = 1.3 \times 10^{-6}\ m$$
- *Range of the wavelength* $\Delta\lambda = 0.17\mu m$:
  $$\lambda_{min} = \lambda - 0.085\ \mu m$$
  $$\lambda_{max} = \lambda + 0.085\ \mu m$$
- *Speed of light* $c = 3 \times 10^{8}\ m/s$

Find: the bandwidth BW for this band.



Ex.2.2 Frequency, Wavelength, and BW

Consider the 1.3-micron band:

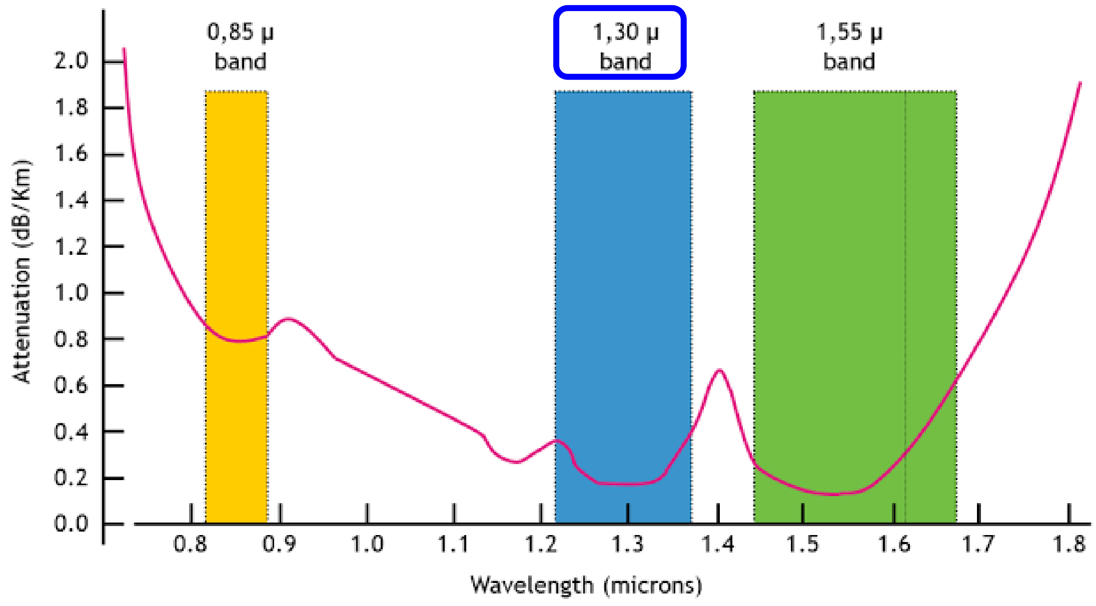
- *Central frequency wavelength*

$$\lambda = 1.3\ \mu m = 1.3 \times 10^{-6}\ m$$

- *Range of the wavelength $\Delta\lambda = 0.17\mu m$:*

$$\lambda_{min} = \lambda - 0.085\ \mu m$$
$$\lambda_{max} = \lambda + 0.085\ \mu m$$

- *Speed of light $c = 3 \times 10^8\ m/s$*



Find: the bandwidth BW for this band.

- *Lowest frequency*

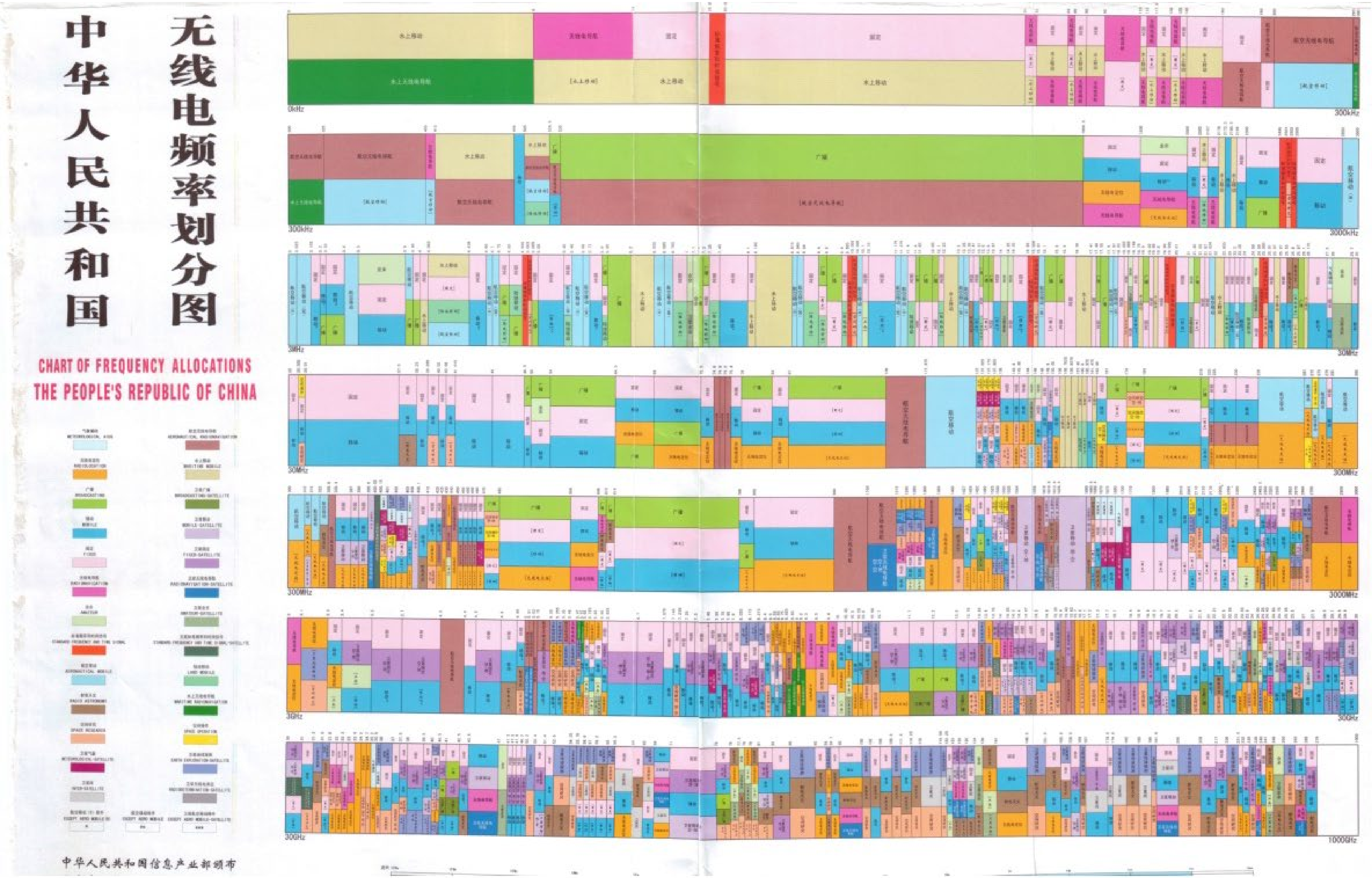
$$f_{min} = \frac{c}{\lambda_{max}} = 2.17 \times 10^{14}\ Hz$$

- *highest frequency*

$$f_{max} = \frac{c}{\lambda_{min}} = 2.47 \times 10^{14}\ Hz$$

- *BW = $f_{max} - f_{min} = 3 \times 10^{13}$Hz = 30 THz*

Fiber communication has a large BW, so it has a huge capacity. -> used in backbone

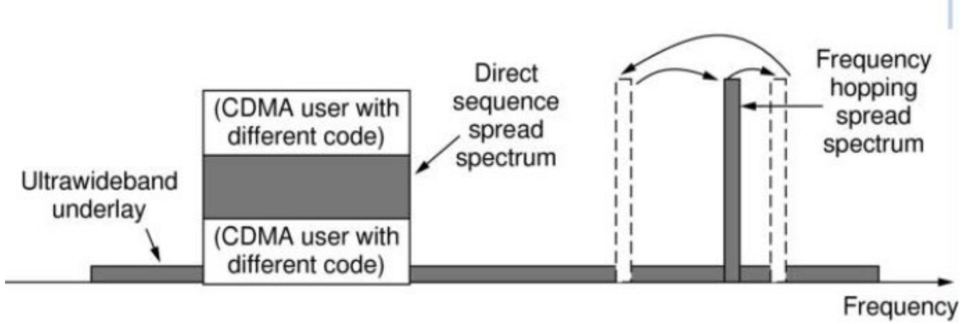


Very crowded

Transmission – Narrow Band v.s. Wide Band

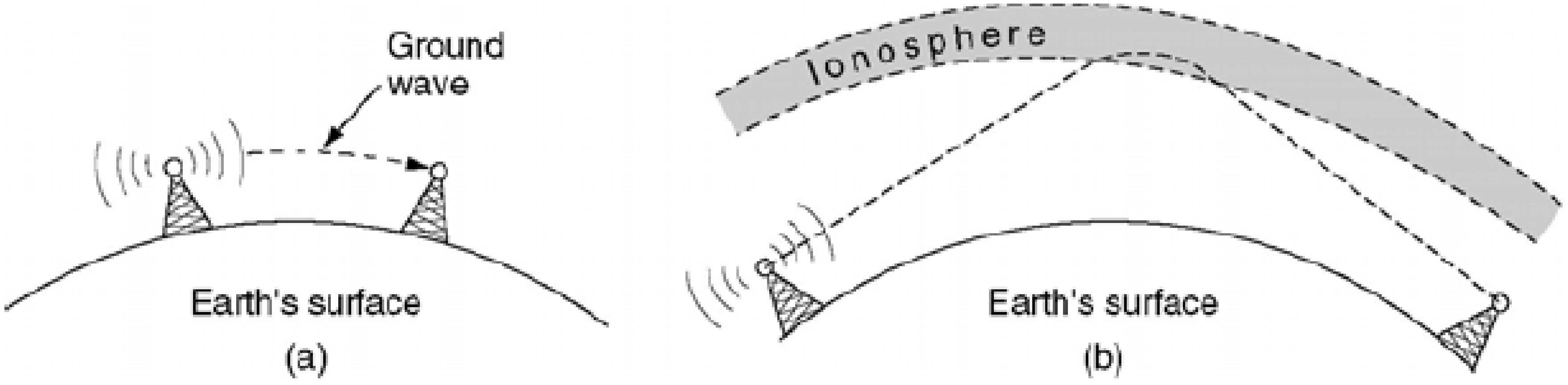
- ***Most transmission use a narrow frequency band.***
  - ***narrow***: *small* $\frac{BW}{f_c}$
  - *Why: good reception*
- ***Wide band transmission (Spread Spectrum) is also widely used.***
  - *Frequency Hopping Spread Spectrum (FHSS)*
    - *Transmitter hops (over 100 times/s) from frequency to frequency. Receiver hops too.*
    - *Hard to detect, almost impossible to jam. E.g. military use.*
  - *Direct Sequence Spread Spectrum (DSSS)*
    - *Popular in mobile telecommunication, e.g. **CDMA**.*
    - *Good spectral efficiency.*

Will learn it today

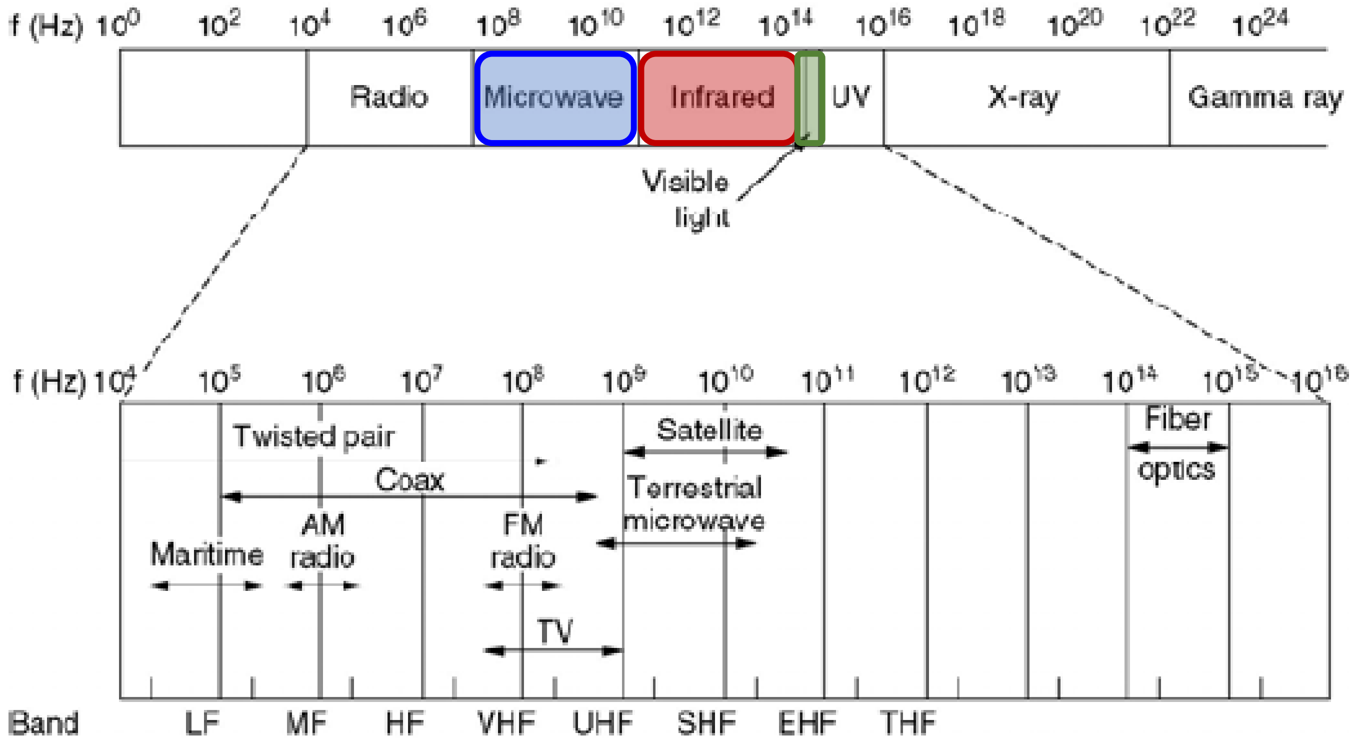


Radio Transmission

- *In VLF (very low frequency), LF (low frequency) and MF (middle frequency) bands, radio waves follow the curvature of the earth.*
- *In HF (high frequency) band, radio waves are reflected by the ionosphere(电离层).*

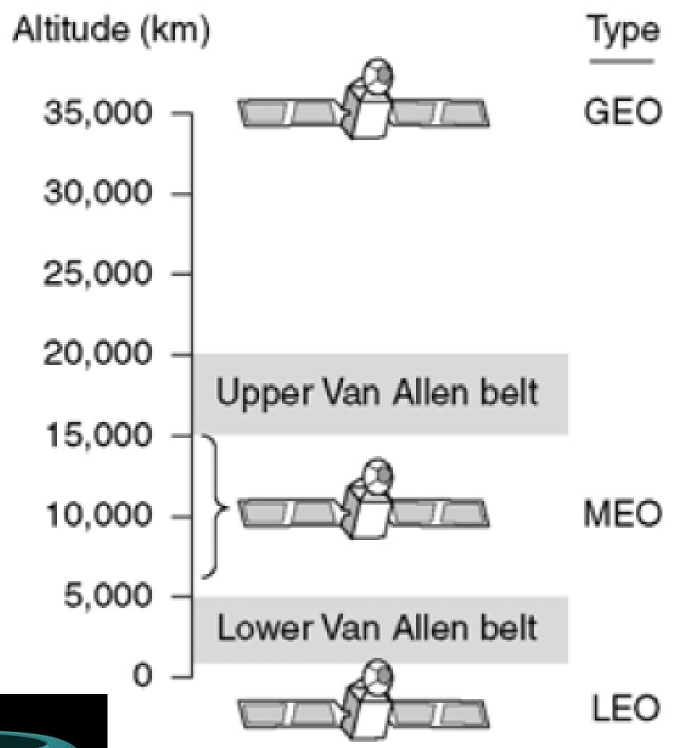


Wireless Transmission



- ***Microwave* (>100MHz)**
  - *Near straight lines*
  - *Antenna high SNR*
  - *Repeaters needed*
  - *Multipath fading*
- ***Infrared*, *mmWave***
  - *Short range*
  - *More like light than radio, which is a good thing.*
- ***Lightwave***
  - *Line-of-sight*

Communication Satellites

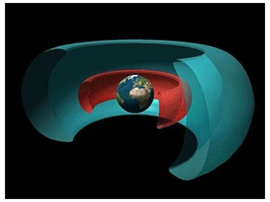


- ***Geostationary***
  - *Not interactive*
- ***Medium-Earth Orbit***
  - *E.g. GPS*
- ***Low-Earth Orbit***
  - *E.g. Starlink*

北斗3

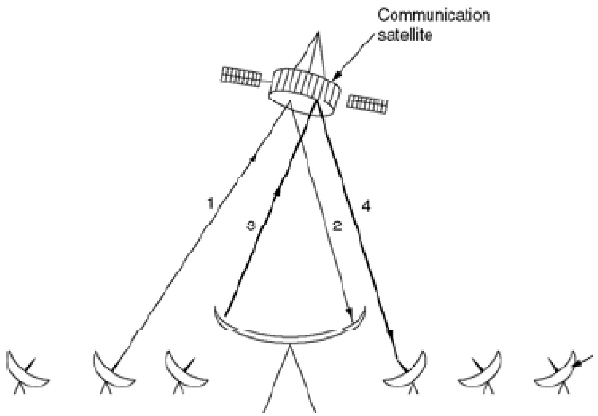
3 GEO

24 MEO



Van Allen Belts

credit: NASA



Summary of satellites, as of 19 May 2023

Block	Launch period	Satellite launches			Currently in orbit and healthy
		Success	Failure	Planned	
1	2000–2006	4	0	0	0
2	2007–2019	20	0	0	15
3	2015–present	36	0	0	31
Total		60	0	0	46

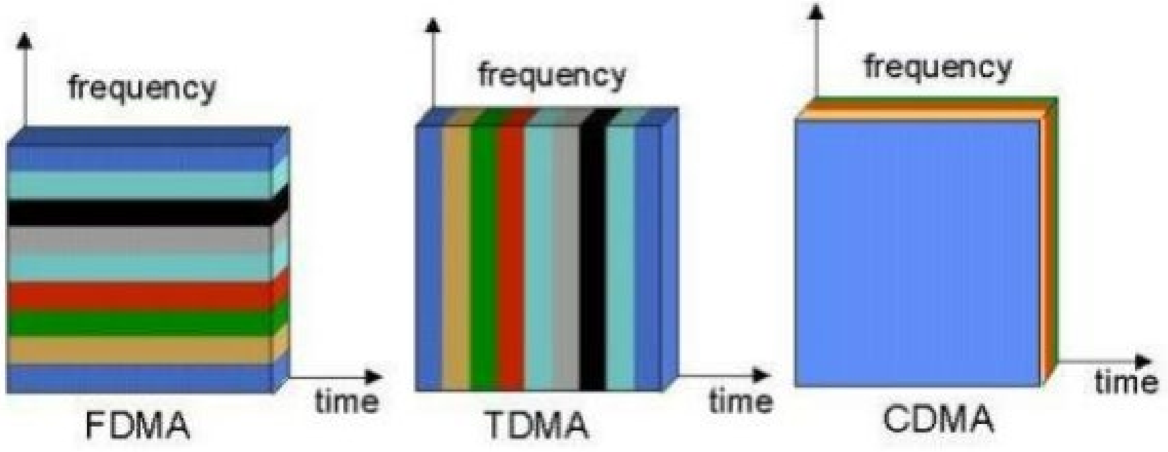
credit: WiKi

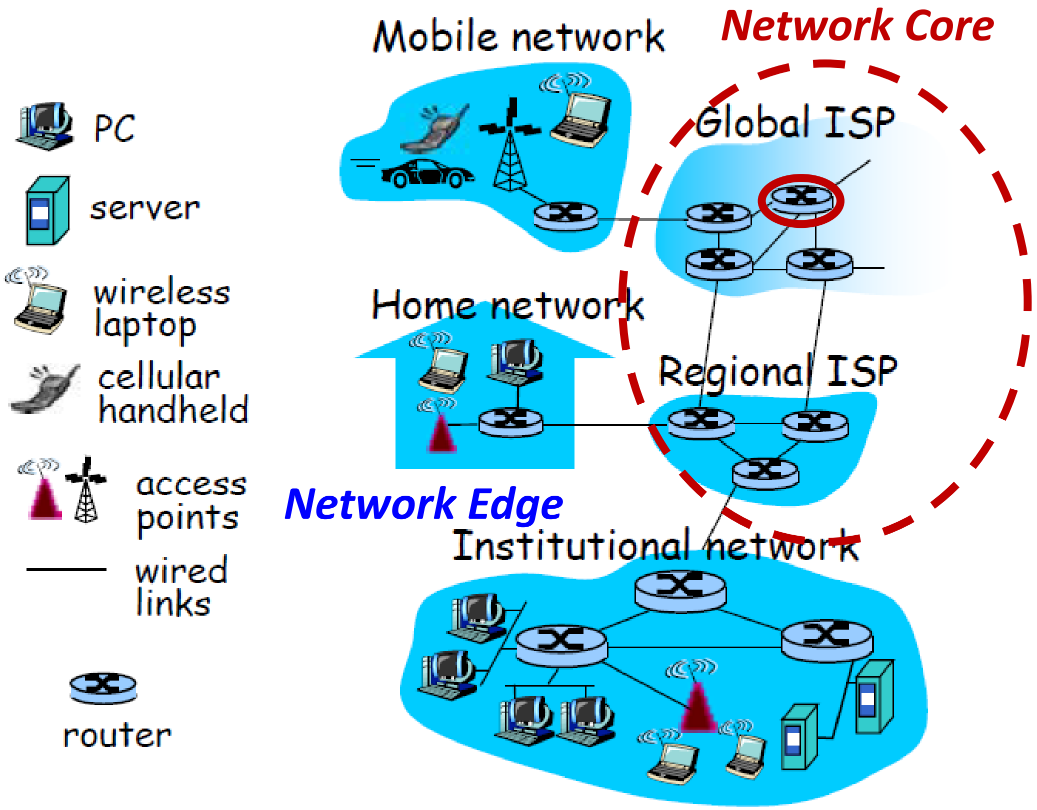
2.4

Multiplexing

Multi-Access

- ***Access Network: Accessing the edge router***
  - *Wired*
  - *Wireless*
- ***Multi-user: multiple access (to channel)***





Multiplexing

- ***Multiplexing (复用)***: *merging multiple flows (of signals) on a single medium, so that multiple users can use a shared medium*
- *What to divide/share:* ***space (S), time (T), frequency (F), code (C)***

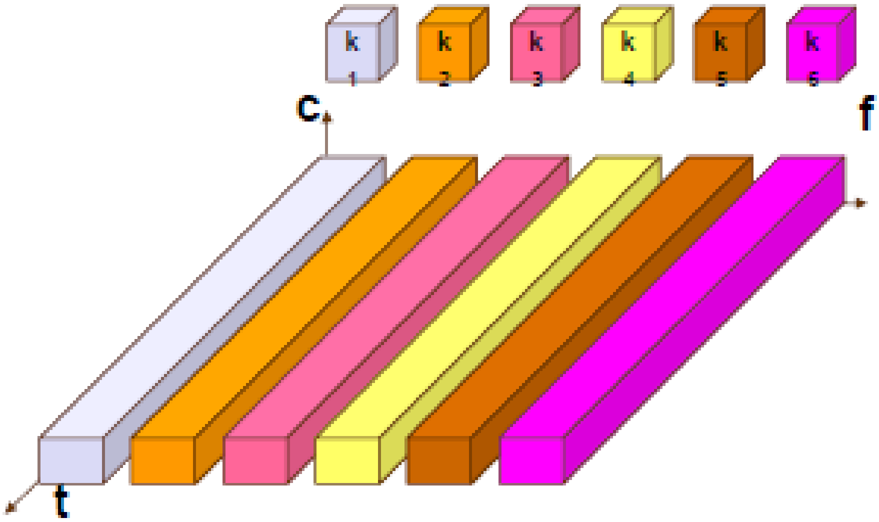
Space Division Multiplexing (SDM) 空分复用

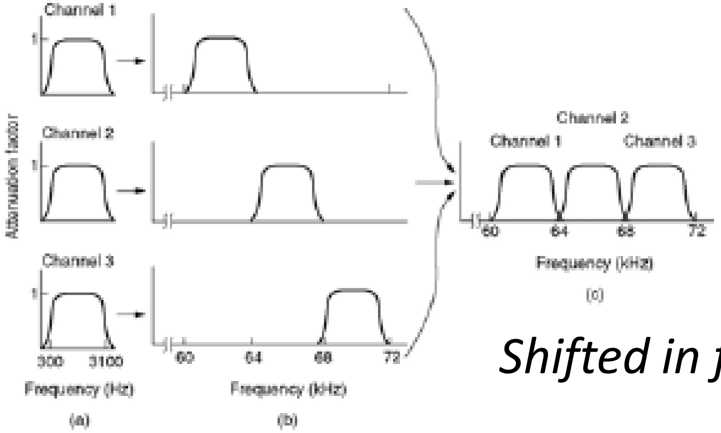
Frequency Division Multiplexing

- ***FDM, FDMA*** **频分复用**
- *The spectrum is divided into smaller frequency bands.*

  e.g. 2.4GHz: ISM bands

- *A channel contains a certain band of spectrum for the entire time.*



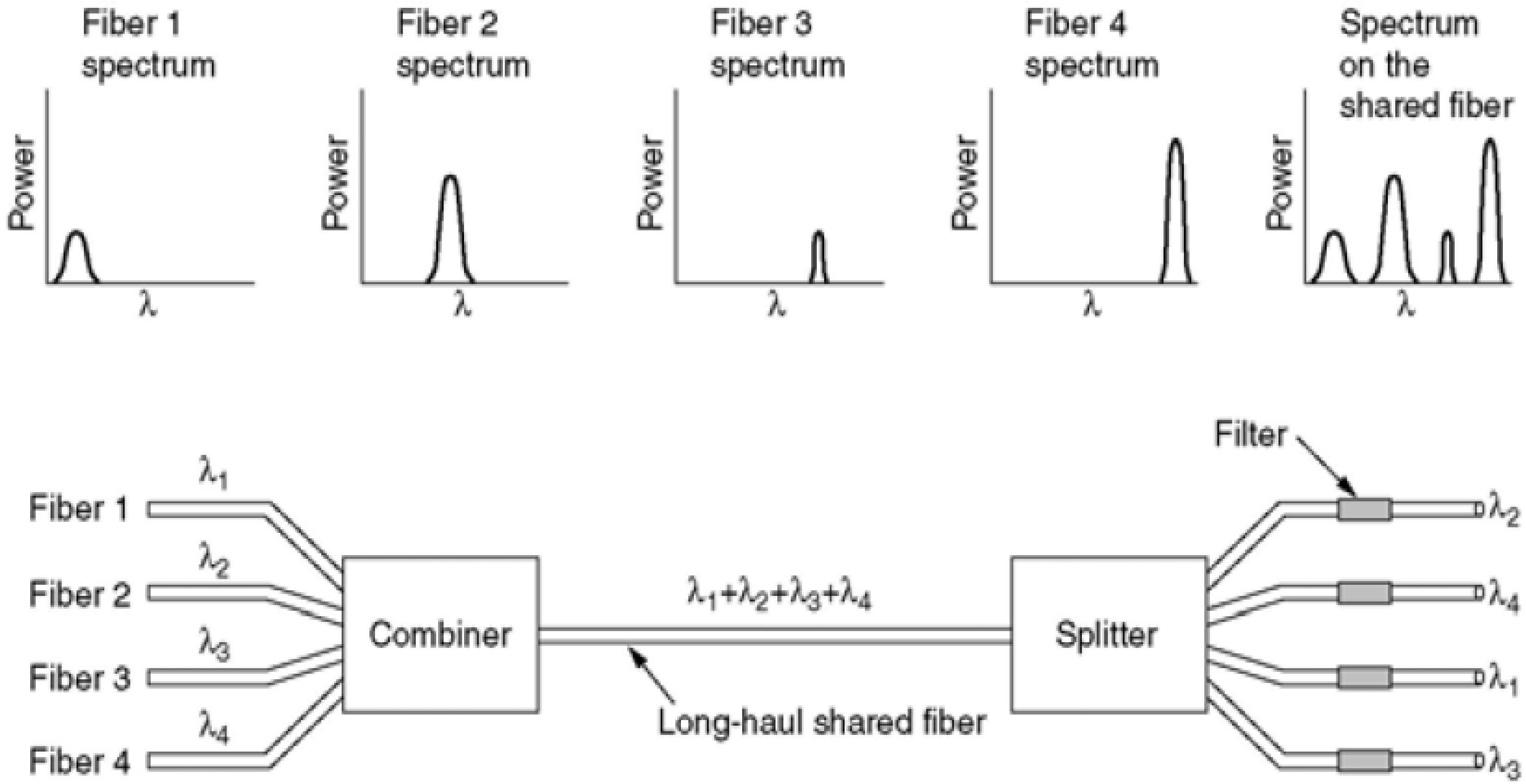


Shifted in frequency

Wavelength Division Multiplexing

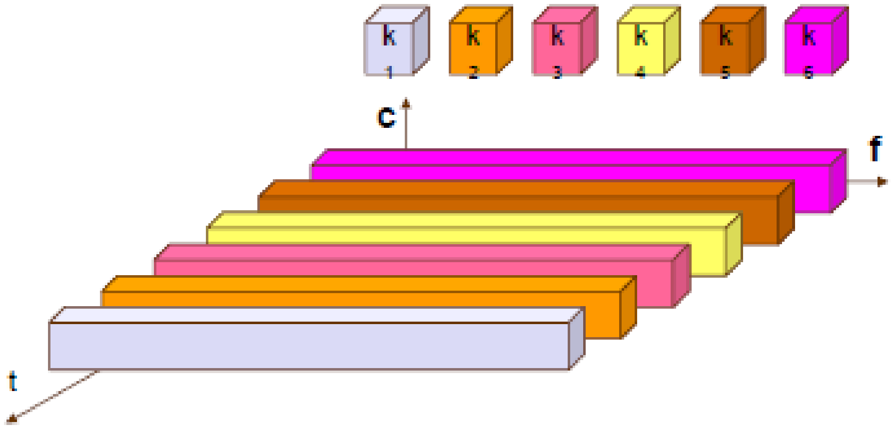
- ***Fundamentally, the same as FDM, but on optics***

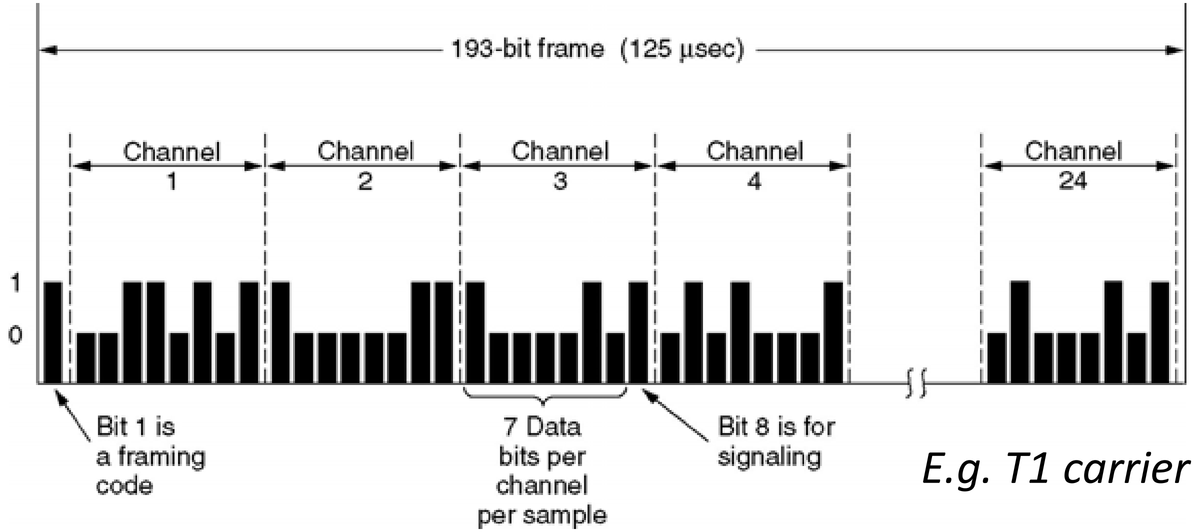
波分复用

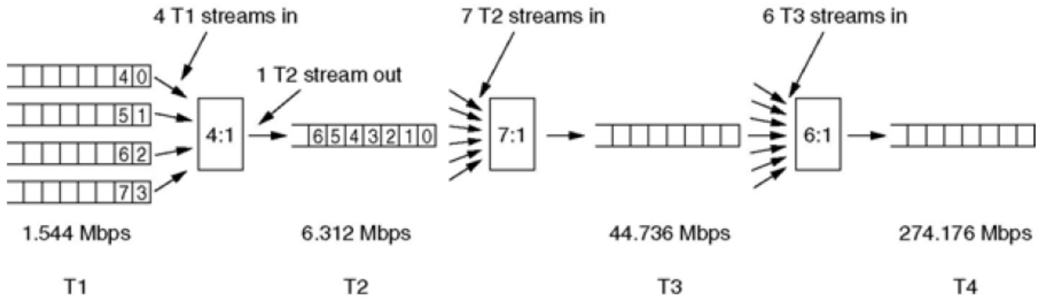


Time Division Multiplexing

- ***TDM, TDMA*** 时分复用
- *A channel uses the entire spectrum for a certain amount of time.*
- *Disadvantage: synchronization (同步)*







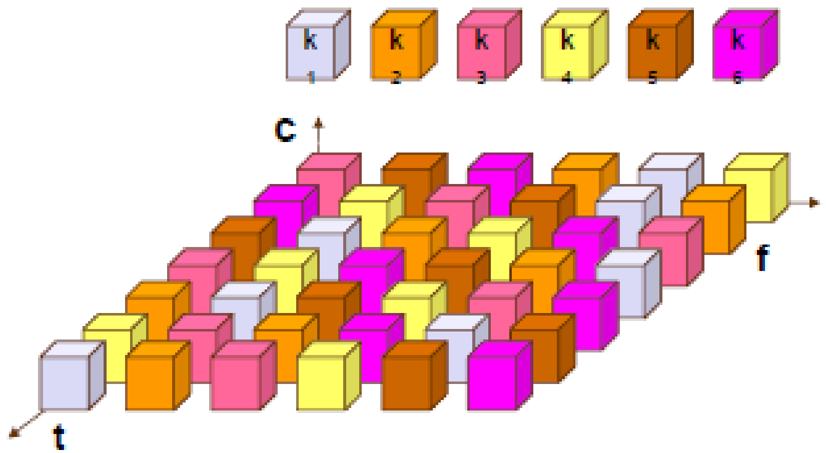
E.g. T1 carrier in telephony networks

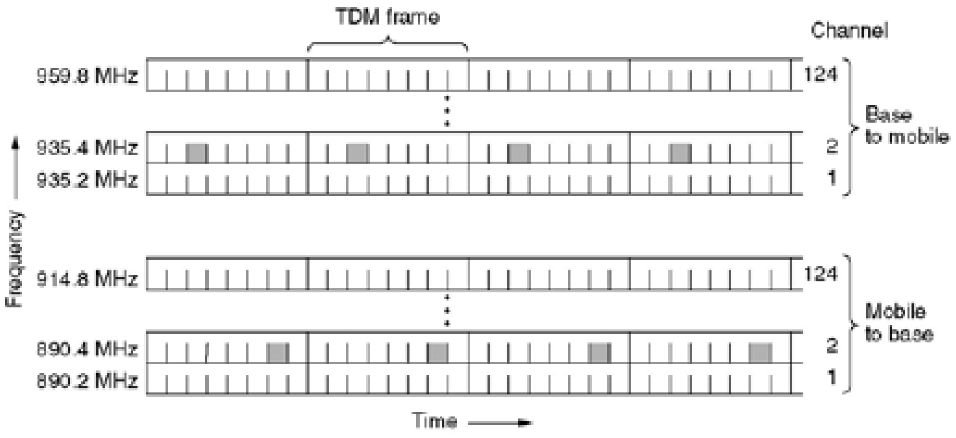
Time and Frequency Multiplexing

- ***Combination of FDM and TDM***
- *A channel uses a certain frequency band for a certain amount of time.*

  E.g. Global System for Mobile Communication (GSM, i.e. 2G)

- *But: needs synchronization (同步), and coordination!*

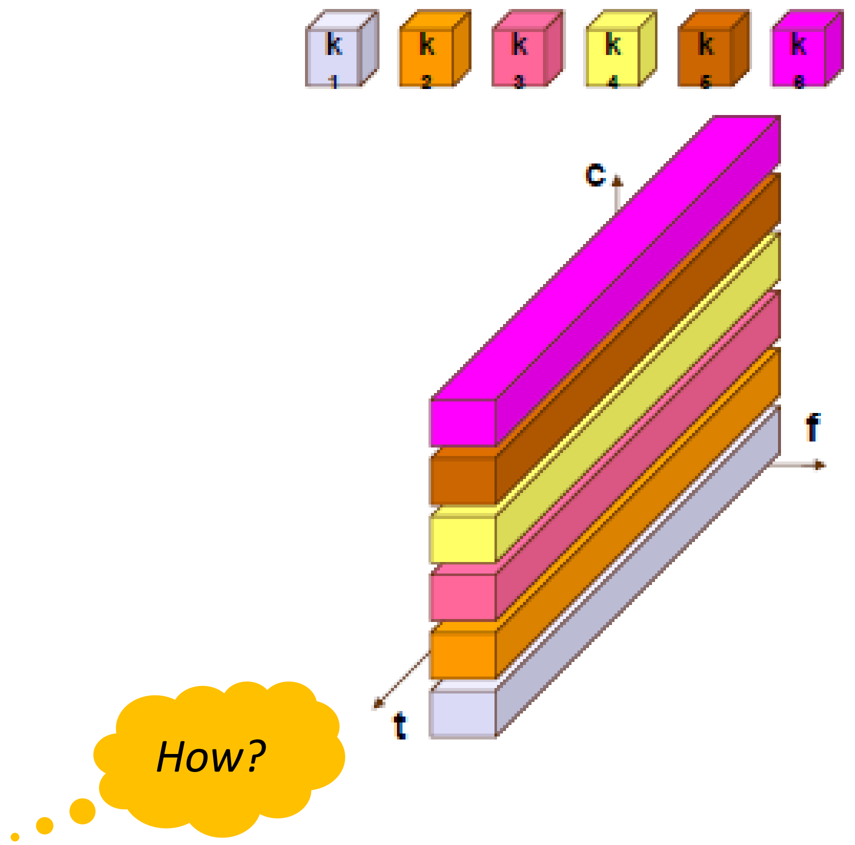




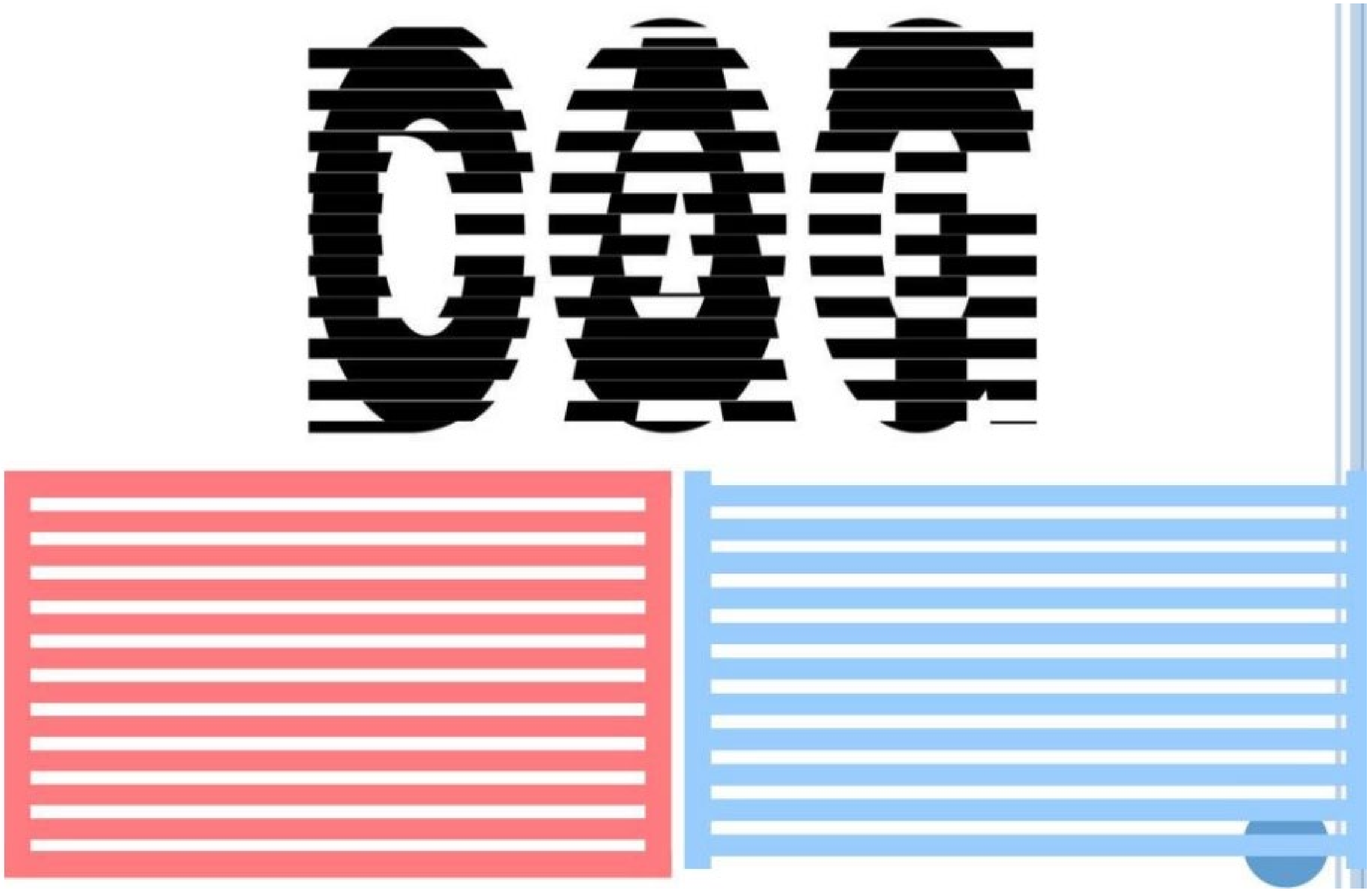
Code Division Multiplexing

- ***CDM and CDMA*** 码分复用
- *Each channel uses the same spectrum at the same time.*

  E.g. CDMA2000, WCDMA (often used with space division multiplexing or FDM)
- *Transmissions are separated using* ***coding theory***.

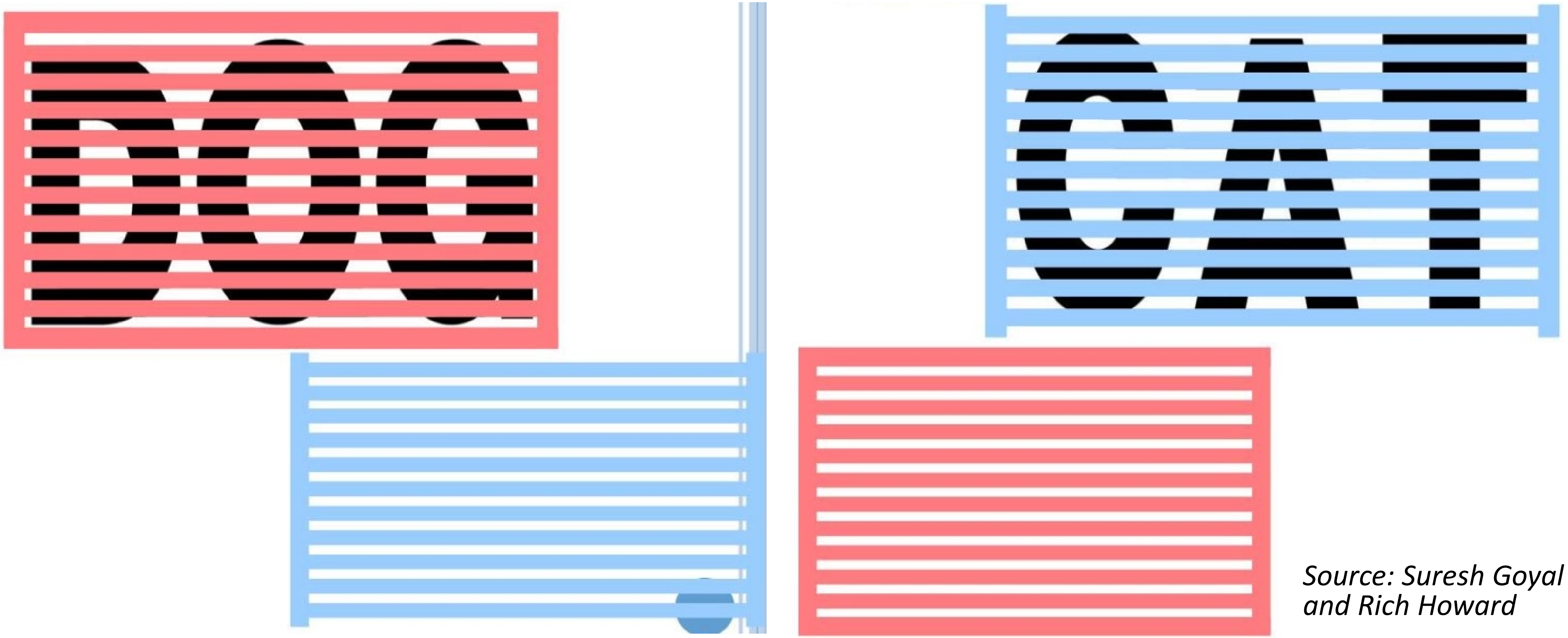


Intuition of CDMA



Source: Suresh Goyal and Rich Howard

Intuition of CDMA



Source: Suresh Goyal and Rich Howard

More on CDMA

- ***Each bit time is divided into m short intervals called chips.***
  - *Usually, **m** = 64 or 128*
- ***Each station A is assigned a unique m-bits code called a chip sequence, A = < A_1, A_2 , … , A_m >***

m-chip vector

1) *Each A_i is either 1 or -1.*

Bipolar

2) *All chip sequences are **pairwise orthogonal (正交)**.*

$$< A, B > = \frac{1}{m}\sum_{i=1}^{m} A_i \cdot B_i = 0$$

3) *Chip sequences are of **unit length**.*

$$< A, A > = \frac{1}{m}\sum_{i=1}^{m} A_i \cdot A_i = 1$$

How does CDMA work?

- ***Three stations: A, B, C***
- ***Chip sequences:***
  - $A = \langle 1\ 1\ {-1}\ {-1} \rangle$
  - $B = \langle 1\ {-1}\ 1\ {-1} \rangle$
  - $C = \langle 1\ {-1}\ {-1}\ 1 \rangle$

Orthogonal

- ***Transmits:***
  - T_A: 1
  - T_B: 0 (i.e. -1)
  - T_C: does not transmit (i.e. 0)
- ***In the air:***

  $S = 1 \cdot A + (-1) \cdot B + 0 \cdot C = ?$

- ***Receivers of A, B, and C:***
  - $R_A : \langle S, A \rangle = 1$

    -> *transmitted 1*
  - $R_B : \langle S, B \rangle = -1$

    -> *transmitted 0*
  - $R_C : \langle S, C \rangle = 0$

    -> *did not transmit*

Ex.2.3 CDMA

Four CDMA Stations and their chip sequences:

A = (−1 −1 −1 +1 +1 −1 +1 +1)

B = (−1 −1 +1 −1 +1 +1 +1 −1)

C = (−1 +1 −1 +1 +1 +1 −1 −1)

D = (−1 +1 −1 −1 −1 −1 +1 −1)

Consider the following cases, calculate the transmitted signal:

1) *C transmits 1, others do not transmit*
2) *A transmitted 1, B transmitted 0, others do not transmit*
3) *A, B, C, D all transmitted 1*

For receiver of C, what can be decoded from the received signal?

Summary of PHY (1/2)

- ***PHY: providing a bit pipe***
- ***Data rate**: the number of bits transmitted per seconds **(bit/s, or bps)***
- ***Signal bandwidth** and the **Nyquist Sampling rule:***
  - *To reconstruct a signal of bandwidth **H** Hz, we need **2H** samples/s*
- ***Channel capacity** for a channel of **BW** bandwidth:*
  - ***Shannon Capacity**: $C = BW\, log_2(1 + \frac{S}{N})$ bps*

- ***Modulation** schemes: digital (bit stream) -> analog (sinusoidal wave)*

$$s(t) = A\cos(2\pi f t + \theta)$$

  - *Amplitude Shifted Keying (ASK), AM*
  - *Frequency Shifted Keying (FSK), FM*
  - *Phase Shifted Keying (PSK), PM – perform well against noise*
  - *Quadratic Amplitude Modulation (QAM): AM + PM*
    - *Constellation diagram*

Summary of PHY (2/2)

- ***Spectrum for wireless communication is crowded***
  - *Frequency and Wavelength* $\lambda \cdot f = c$
  - *Differentiate:* ***Channel BW*** *v.s. Carrier Frequency f (bands)*
- ***Multiplexing****: mixing flows of different users onto the same medium*
  - ***SDM, TDM, FDM, WDM***

    Each stands for what?
  - ***CDMA***
    - *Direct Sequence Spread Spectrum (DSSS)*
    - *Large capacity, resistant against noise*
    - ***Chip sequence (Rule 1,2,3): encoding and decoding***